CSS 426 MLOPS

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BATCH : 1

ASSIGNMENT NO. : 1

PART : 1

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1. What breaks (or becomes harder) when ML is added?

The introduction of machine learning fundamentally changes the system from a deterministic, rule-driven service to a probabilistic, data-driven system. This transition introduces several technical and operational complexities that must be addressed during system design and deployment.

- First, the system loses its fully deterministic nature. In a rule-based system, each output is directly tied to explicitly defined business logic, making the reasoning process transparent and easily traceable. In contrast, an ML model produces predictions based on learned statistical relationships between features and outcomes. While this enables the system to capture complex patterns, it also reduces transparency, making it more difficult to directly explain or validate individual decisions without additional interpretability mechanisms.
- Second, data quality becomes a critical dependency. In a rule-based system, incorrect or missing data typically affects only specific rule evaluations. However, in an ML-based system, the model relies on consistent and properly processed feature inputs. Issues such as missing values, inconsistent formats, incorrect encoding, or changes in feature distributions can significantly affect prediction accuracy. This makes data validation and feature consistency essential components of the system.
- Third, the system architecture becomes more complex due to the separation between training and inference processes. The model must first be trained offline using historical data and then deployed for online prediction. This separation introduces the risk of inconsistencies between training and production environments, particularly if feature engineering logic is implemented differently in each environment.
- Fourth, testing requirements expand beyond traditional software validation. While unit tests and integration tests can verify API functionality and system stability, they cannot guarantee model correctness in terms of prediction quality. ML systems must be evaluated using statistical

performance metrics such as F1 score, ROC-AUC, and Precision-Recall. These metrics must be tracked and compared across model versions to ensure reliability.

- Finally, system maintenance becomes more complex. In a rule-based system, behavior changes only when business rules are modified. In contrast, ML model performance may degrade over time due to changes in customer behavior, support patterns, or business conditions. This introduces the need for ongoing monitoring, performance evaluation, and periodic retraining.

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2. Why DevOps alone is insufficient?

DevOps practices focus on ensuring reliable software delivery, infrastructure automation, and operational stability. While these practices are essential, they are not sufficient for managing ML systems because ML introduces dependencies on data and model performance, not just application code. DevOps pipelines ensure that code is correctly built, tested, and deployed, and they monitor service-level metrics such as availability, latency, and error rates. However, these metrics do not reflect the accuracy or effectiveness of ML predictions. A deployed model may continue to operate normally from an infrastructure perspective while producing degraded or incorrect predictions. ML systems require additional lifecycle management that is not addressed by traditional DevOps processes. This includes tracking training datasets, maintaining model version history, validating feature consistency, and ensuring reproducibility. Without proper model and data versioning, it becomes difficult to determine which model version is deployed, what data it was trained on, and how it performs relative to previous versions. Furthermore, ML systems must handle changes in input data distributions and relationships between features and outcomes. Traditional DevOps monitoring tools are designed to detect infrastructure and application failures, but they do not monitor data drift, model performance degradation, or prediction reliability. ML systems also require structured model deployment and validation workflows. New models must be evaluated, compared against existing models, and safely deployed using controlled release strategies. These requirements extend beyond the scope of conventional DevOps and require specialized MLOps practices.

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3. What new risks does ML introduce?

Machine learning introduces several new risks that are not present, or are significantly less prominent, in rule-based systems.

- One major risk is data drift, where the statistical properties of input data change over time. For example, changes in customer behavior, service usage patterns, or support processes may

result in input data distributions that differ from those used during training. This can reduce the model's ability to make accurate predictions.

- Another important risk is concept drift, where the relationship between input features and the target outcome changes over time. A feature that was previously a strong indicator of churn may become less relevant due to business changes, customer retention strategies, or product improvements. This can reduce the predictive validity of the model.
- Training–serving inconsistency is another potential risk. If feature engineering processes differ between training and inference environments, the model may receive inputs in a different format or distribution than expected, resulting in incorrect predictions.

ML systems also introduce risks related to performance degradation. Even if a model performs well during initial evaluation, its performance may decline over time as underlying conditions change. Without proper monitoring and retraining strategies, such degradation may remain undetected Reproducibility and version control represent additional risks. ML models depend on specific datasets, preprocessing steps, and training configurations. Without proper version management, it becomes difficult to reproduce results, audit system behavior, or investigate prediction outcomes. Operational risks also increase due to additional dependencies such as model artifacts, feature encoders, and ML libraries. Compatibility issues or incorrect configurations can disrupt inference services. Finally, ML systems introduce explainability and governance challenges. Unlike rule-based systems, model predictions are derived from learned statistical relationships rather than explicitly defined logic. This can make it more difficult to interpret decisions, justify outcomes, and maintain stakeholder confidence.

Dataset Description: Telco Customer Churn Dataset

The Telco Customer Churn dataset from Kaggle is used as the primary data source for this assignment. This dataset contains structured information about telecom customers, their service subscriptions, account details, billing information, and churn status. It is widely used for churn prediction and customer behavior analysis. The dataset consists of 7,043 customer records and 21 columns. Each record represents an individual customer and contains the following attributes:

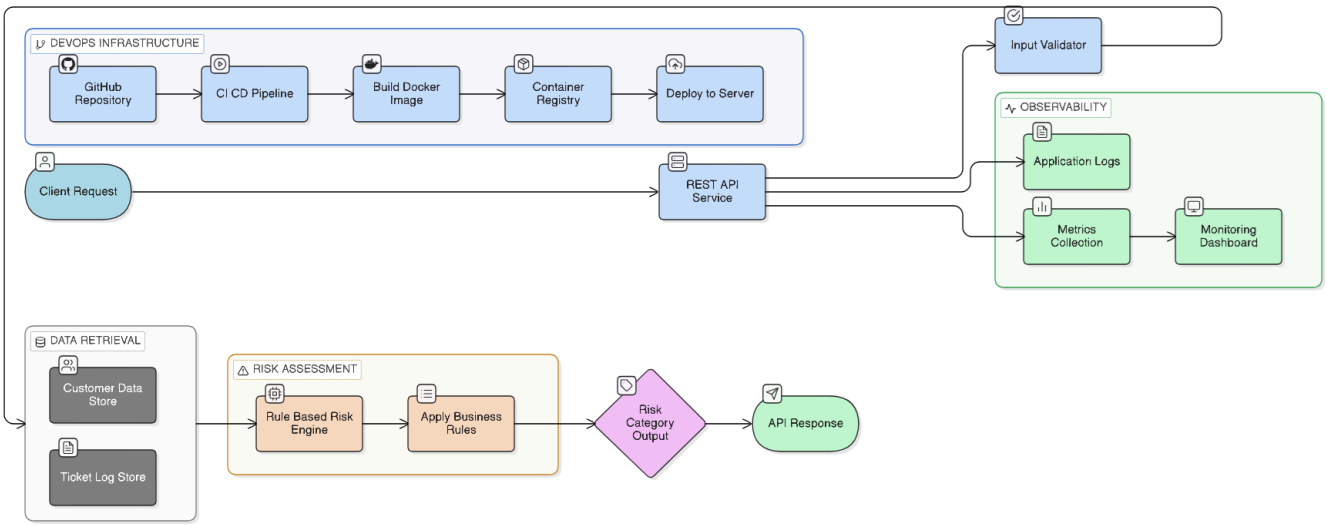
1. **customerID** – Unique identifier assigned to each customer
2. **gender** – Indicates whether the customer is male or female
3. **SeniorCitizen** – Indicates whether the customer is a senior citizen (0 or 1)
4. **Partner** – Indicates whether the customer has a partner
5. **Dependents** – Indicates whether the customer has dependents
6. **tenure** – Number of months the customer has stayed with the company
7. **PhoneService** – Indicates whether the customer has phone service
8. **MultipleLines** – Indicates whether the customer has multiple phone lines
9. **InternetService** – Type of internet service (DSL, Fiber optic, or No service)

10. **OnlineSecurity** – Indicates whether online security service is enabled
11. **OnlineBackup** – Indicates whether online backup service is enabled
12. **DeviceProtection** – Indicates whether device protection service is enabled
13. **TechSupport** – Indicates whether technical support service is enabled
14. **StreamingTV** – Indicates whether streaming TV service is enabled
15. **StreamingMovies** – Indicates whether streaming movie service is enabled
16. **Contract** – Type of contract (Month-to-month, One year, or Two year)
17. **PaperlessBilling** – Indicates whether paperless billing is enabled
18. **PaymentMethod** – Method used for payment
19. **MonthlyCharges** – Monthly amount charged to the customer
20. **TotalCharges** – Total amount charged to the customer
21. **Churn** – Target variable indicating whether the customer has churned (Yes or No)

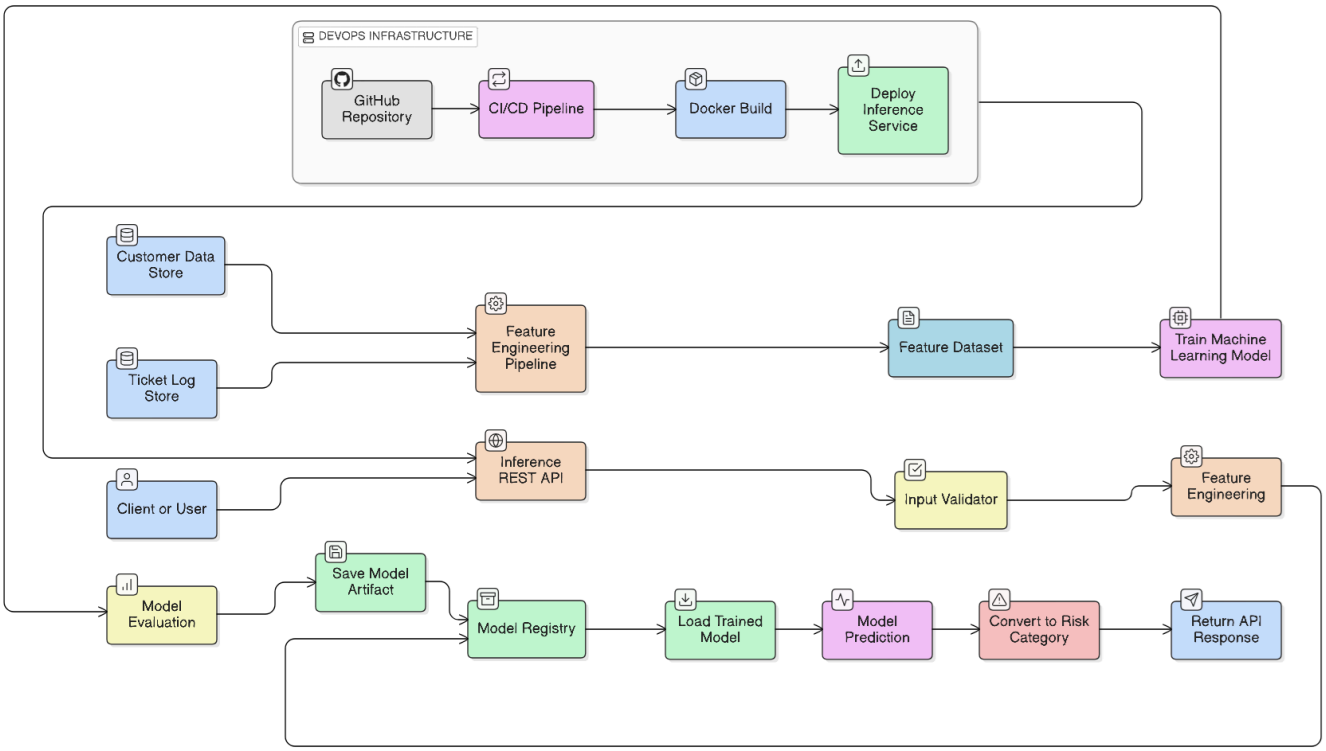
The dataset contains both categorical and numerical attributes that describe customer demographics, service usage, and billing behavior. The Churn column serves as the target variable and indicates whether a customer has discontinued the telecom service. This dataset provides a structured foundation for analyzing customer retention and churn patterns.

Link :- https://www.kaggle.com/datasets/blastchar/telco-customer-churn

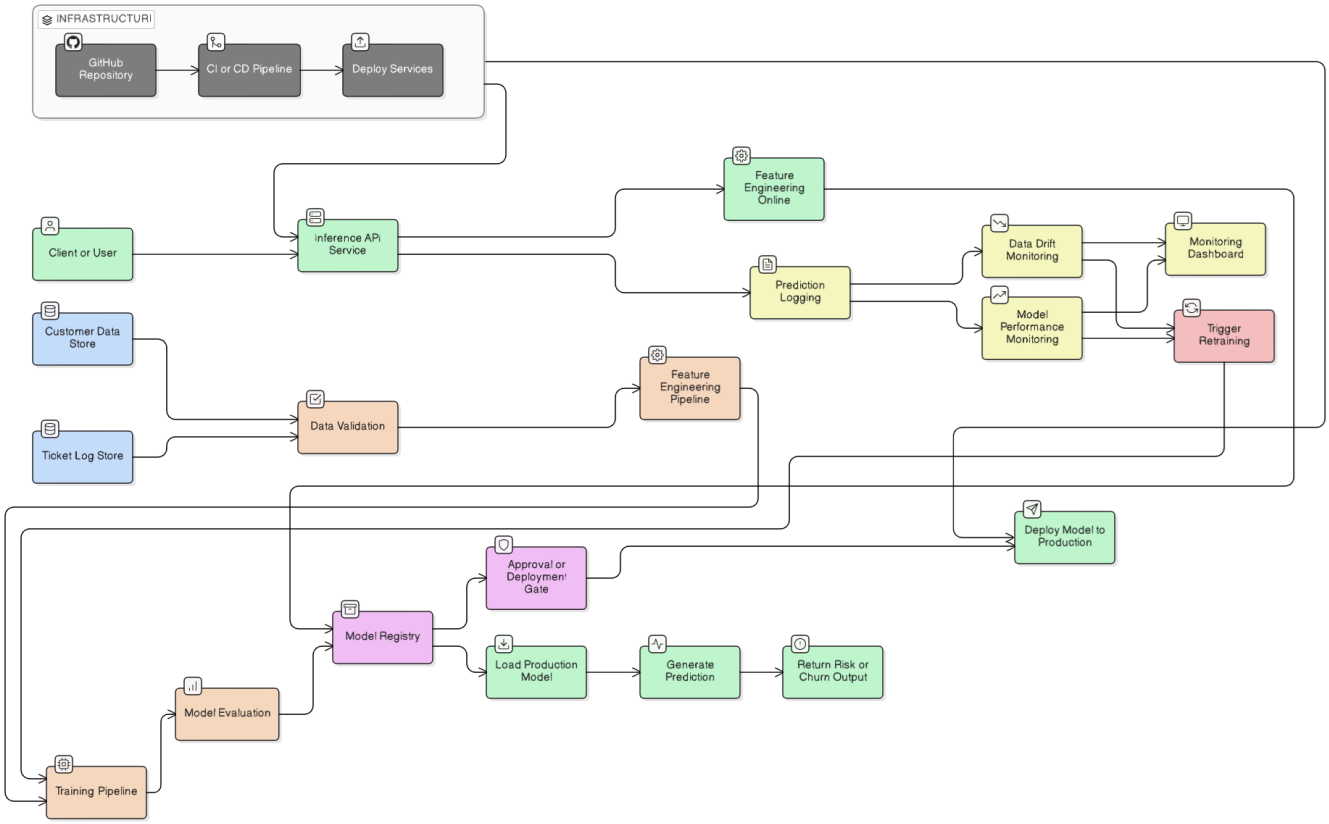
Workflow for DevOps System



Workflow for ML System



Workflow for MLOps System


INFRASTRUCTURI
GitHub Repository
CI or CD Pipeline
Deploy Services
Client or User
Inference API Service
Feature Engineering Online
Prediction Logging
Data Drift Monitoring
Monitoring Dashboard
Model Performance Monitoring
Trigger Retraining
Customer Data Store
Ticket Log Store
Data Validation
Feature Engineering Pipeline
Deploy Model to Production
Approval or Deployment Gate
Model Registry
Load Production Model
Generate Prediction
Return Risk or Churn Output
Model Evaluation
Training Pipeline